

LAYLA LIU

732-546-3198

liulayla.490ll@outlook.com

SUMMARY:

- Highly motivated and data-oriented Big Data Developer/Engineer with working experience in the domains including financial, telecommunication, and E-Commerce.
- Solid understanding of **Hadoop Ecosystem (3.0)** components such as **HDFS, MapReduce, YARN, ZooKeeper (3.4), Spark (2&3), Kafka(0.11), HBase (2.x), Hive (2.x), Pig, Sqoop (1.4), Flume(1.6), Flink(1.10), Oozie, Phoenix, NiFi, Impala**, etc.
- Solid programming skills in both OOP and functional languages such as **JAVA, Python, Scala**, and **SQL, R**.
- Proficiency in writing different SQL queries such as **MySQL, T-SQL, PL/SQL, Pig Latin**, and query analysis framework including **Spark SQL, Hive, Flink SQL, Pig**
- Experience in all the phases of Big Data Application life cycle involving requirement analysis, design, coding, testing, and deployment
- Design, create, build & maintain data pipelines and implement massive **ETL (Extract, Transform, Load)/ Data Warehousing** workflow using **SSIS, Talend** and **Informatica**
- Knowledge of **Big Data Testing**, performance optimization, capacity planning, monitoring and performance tuning of **Hadoop clusters**
- Familiar with scheduling tools including **Oozie, Azkaban** and cluster monitoring tools such as **Ambari, Cloudera Manager**
- Experience in building complete infrastructure to ingest (**Sqoop, Flume**), transform (**Hive, Tez, PySpark**) & store data for further analysis & business requirement.
- Working experience on real-time data processing with **Spark Streaming** and **Kafka** in cloud computing, and batch processing framework including **MapReduce, Spark**, and **Flink**
- Deep understanding about **EDW architecture (Snowflake schema & Star schema)**, data modeling (**OLTP & OLAP**) schema, and **Data Lake (S3, Snowflake)**.
- Experience in working with **RDBMS** including **Oracle, MS SQL** and **MySQL (8.x)**.
- Experience in developing scalable solutions using **NoSQL** databases including **Redis, Cassandra(2.1), HBase (2.x)** and **MongoDB (3.6)**
- Work experience with cloud storage and computing on **AWS (EC2, S3, EMR, RedShift, Glue, Athena, Lambda, Kinesis, DynamoDB, Glue)**
- Strong ability to implement statistical methods (**A/B testing, ANOVA, Hypothesis Testing, PCA, time series analysis, linear/logistic regression, Design of Experiment**) using **R, Python, MATLAB**, and **SAS** for exploring hidden patterns in the data.
- Adapt in applying machine learning techniques (**classification, clustering, decision tree, random forests, regression trees, KNN, NLP, SVM, neural network, deep learning**) using **R, Python, PySpark** and **Spark MLlib** to fulfill business needs
- Knowledge of data serialization and familiar with data formats including **Sequence File, Avro, Parquet, XML, ORC** and **JSON**
- Strong in **Core Java, Data Structure and Algorithms**, and **Object-Oriented Design**
- Familiar with various technologies such as **Java EE, Maven, Elastic- Search, Splunk, Akka**
- Demonstrated ability to create visually appealing and interactive reports/dashboards using software such as **Tableau, Jupyter Notebook, Zeppelin**, etc.

- Expose to various software development methodologies including **Agile**, **Scrum**, etc. by using version control tools (**Git**), CI/CD tools (**Jenkins**), and management tools (**JIRA**)
- Good team player but can also work independently in a fast-moving environment

TECHNICAL SKILLS:

Hadoop/Spark Ecosystem: \

Hadoop 3.x, MapReduce, Spark 2&3, Pig 0.17, \
Hive 2.x, Sqoop 1.4.7, Flume 1.6.0, Kafka, \ 2.1.x, MongoDB 3.6, Microsoft SQL Server 12.0\
Yarn, Oozie, Zookeeper 3.4.x, Flink 1.10\

Programming Language: \

Java 8.x, Scala 2.12, Python 3, SQL, Unix/\
Bash shell, R 3.6.0, SAS 9.4 \

Cloud Platform: \

Amazon Web Services: EC2/EMR/S3 \
RedShift, Glue, Athena, Lambda, RDS, \
DynamoDB\

Packages \

Python - Pandas, Numpy, SciKit-learn Seaborn, \
Matplotlib, SciPy, Scrapy, Tensorflow; \
R- e1071, dplyr, ggplot2, gmodels, stats, \
caret; NLTK, MLlib\

Tools: Tableau, Microsoft Office (Word, Excel with macros, PowerPoint), Jupyter Notebook, SPSS, SAS, Putty, MATLAB, Azure ML, Minitab, Google Analytics/Ads\

Database: \

Oracle 11g, MySQL 8.x, HBase 2.x, Cassandra \
2.1.x, MongoDB 3.6, Microsoft SQL Server 12.0\

Operating System: \

Unix/Linux, Mac OS, Windows \

Environment & Tools: \

Git, Agile/Scrum, Jenkins, JIRA, IntelliJ IDEA, \
Eclipse, Cloudera (CDH), Hortonworks (HDP) \

Machine Learning Algorithms\

Naïve Bayes, Decision tree, Linear and Logistic \
Regression, SVM, KNN, Random Forests,
k-means, Kohonen Nets, Gradient Boosting, \
Time-Series, Text Mining \

PROFESSIONAL EXPERIENCE:

Marlabs Inc, Piscataway, NJ

02/2020 – Till Date

Client: Samsung SDS America Inc

Role: Big Data Developer

Project: User Behavior Analysis

Description:

This project was mainly used in Internet e-commerce enterprises, using the big data statistical analysis platform developed by Spark and Flink technology to conduct a complex analysis of various user behaviors (access behaviors, shopping behaviors, advertising click behaviors, etc.) of e-commerce websites. Used the statistically analyzed data to assist PMs (product managers), data analysts, and managers in the company to analyze the status of existing products, and continuously improve product design based on user behavior analysis results and adjust the company's strategy and business.

Responsibilities:

- Developed and optimized multi-thread scripts using **Flume** and **Kafka** producer and consumer
- Deployed **Hadoop** Applications on a persistent **Elastic MapReduce (EMR)** cluster through **S3**

- Configured **Amazon S3** and **DynamoDB** as **Kafka** consumer to ingest collected web server logs and data from other sources
- Developed **UDF, UDAF, UDTF** in **Spark SQL** to process, transform, aggregate data in ETL
- Calculated offline new user's retention rate and top 10 popular products using **Spark Core** and **Spark SQL**.
- Consumed streaming data from **Kafka** cluster by **Flink** and used **Flink** to clean the log data
- Integrated **Spark Streaming** and **Flink** by **Scala** to compute real-time statistics on sales of each state and construct user preference analysis.
- Used **machine learning** model such as **Random Forests, Support Vector Machine**, etc. to classify the client's behavior sequence in **Spark MLlib** and **PySpark**.
- Used **Oozie** for scheduling of offline statistical tasks and used **Zookeeper** to monitor the Hadoop cluster and Message Queue of **Kafka**
- Wrote **Shell** script to maintain ETL process and converted raw data into **JSON**.
- Manipulate testing on data pipelines and worked on debug, troubleshooting the issue arise such as data skew, data pipeline stuck, data missing.
- Populated previously calculated offline and real-time statistics into **AWS Redshift** for BI team.

Environments:

Hadoop 3.0, Apache Spark 2, Flink, Flume, Kafka, HDFS, MongoDB, Oozie, Zookeeper, SQL, Java, Linux Shell, Scala 2.12, AWS S3, EMR, DynamoDB, Redshift

Red Line Material Group Data Warehouse

04/2019 – 02/2020

Role: Big Data Developer

Tempe, AZ

Description:

This project helped to build a customer behavior acquisition platform, so it requires to build large scale fault tolerant data collection and processing pipelines from the ground up. This work included complete offline and real-time data warehouse construction and provide data integration, data ingestion, data enrichment/transformation and data modeling from various sources.

Responsibilities:

- Contributed in designing, developing, and maintaining **Data Warehouse** and ETL pipelines
- Built data pipeline for data ingestion and consumed by two system: Offline-processing system and Streaming-processing system.
- Worked with **Sqoop** import and export functionalities to handle large data set transfer between **RDBMS (MySQL)** database and **HBase**
- Configured clusters on **AWS EMR, S3** and maintain data pipelines integrated with **Hive, Spark** and **HBase**
- Built big data solutions using **Phoenix** and **HBase** handling millions of records for the different trends of data and exporting it to **Hive**
- Implemented **Partitioning, Bucketing** in **Hive** for better organization of the data.
- Implemented Hive custom **UDFs** and Analyzed large data sets by **Java** to achieve comprehensive data analysis
- Utilized **Flume** to collect log files and sink the data into **Kafka clusters** and store in **S3**
- Loaded and transformed large sets of structured data by **AWS Athena** and **Glue**

- Enhanced and optimized product Spark code to aggregate, group and run data mining tasks using Spark framework such as **Spark Core, Spark SQL, DataFrame API, Pair RDD's, Spark YARN**
- Processed data by **Spark Streaming** for ingest into **AWS ElasticSearch** to support search and analytics use cases
- Developed Spark application scripts by **Scala** command in **Zeppelin**
- Used Shell scripting to analyze the data from various sources and processed it to store it into **S3** and **Snowflake**.
- Involved in managing nodes on **Hadoop cluster** and monitor Hadoop cluster job performance
- using **Ambari** and send alert by email
- Monitored workflows and scheduling by **Azkaban** and **Informatica**
- Manipulate big data testing such as data staging validation, performance testing, etc. and troubleshooting including data skew, etc.
- Create data engineering diagram and interactive dashboards by **Tableau** and **Lucidcharts**
- Used **Git** for version control and **JIRA** for project tracking.

Environments:

Hadoop 3, Kafka, Sqoop, HDFS, HBase, Hive, Phoenix, Flume, Spark 2, Scala, Java, SQL, MySQL 8.x, Azkaban, Git, JIRA, Tableau, Snowflake, AWS EMR, S3. Athena, Glue

City of Phoenix, Phoenix, AZ

11/2018 – 04/2019

Role: Data Engineer

Project: Data Analysis and engineering

Responsibilities:

- Involved ETL processes including data cleaning, data processing and data storage using **NiFi, SQL Server, Cassandra**.
- Constructed (**Exploratory Data Analysis**) **EDA** to analyze the pattern and trends of regions and get insights of data by **pandas, NumPy, & seaborn** in **Python** and **PySpark**
- Used **Machine Learning algorithm** to explore models including **Decision Tree, SVM** and **Random Forest** to train dataset and tuned parameter by **Spark MLlib, Spark SQL**
- Built up model of image recognition to help stakeholder more accurately and efficiently identify containers with contamination over 20% in Pytorch, and tracked and evaluated the residents' recycling behavior
- Collaborate and tracking the work with **Git**.
- Created ad-hoc reports and presented to clients with customer-friendly **Tableau** dashboards and insightful slides

Environments:

Sqoop, SQL Server, NiFi, Cassandra, Git, Tableau, Python, PySpark

Douman E-commerce Recommendation System

08/2017 – 10/2018

Role: Big Data Engineer

Tempe, AZ

Responsibilities:

- Used **Pig Latin** to do process to the log data, wrote the cleaned structured and semi structured data into numerical tables and saved them into **Hive** as **Parque**
- Developed optimized and scalable machine learning algorithms in **Scala** and **Python** capable of performing predictive modelling using **Spark**, **SparkML**, **Scikit-learn** and other packages
- Developed and implemented offline recommendation algorithms such as **Alternating Least Squares (ALS)** and **Item-CF** using **Spark MLlib**
- Used **Flink API** to develop user profile analysis by Java programming and sink the data into **Oracle**.
- Responsible for building the code dependencies using Maven
- Implemented website interface of near real-time data analytics and visualization using **Spring MVC** by **Java**
- Used **IntelliJ** as Integrated Development Environments
- Used Multidimensional scaling method to construct brand perceptual map to identify competitors in E-commerce market in **R**

Environments:

Apache Pig, Spark, Hive, R, Python, Java, Oracle, IntelliJ

Cheetah Mobile call log analysis

10/2016 - 05/2017

Role: Data Engineer

Seattle, WA

Responsibilities:

- Used **NiFi** to ingest data from various data sources to NoSQL database/ **MongoDB**.
- Collect Call Log files from **AWS Kinesis Stream** by **AWS Lambda**
- Deployed **MapReduce** jobs by Java to process the data from **MongoDB**
- Preprocessed text by lowering case, removing digits and punctuations, removing stop words and short words, retaining only unique words, and stemming the words in Python to reduce the text size by 70%
- Created new concept category and made sentimental analysis on customers' comments in **SPSS**
- Used Kohonen net clustering to improve prediction of customer churn, and build keyword hierarchy by Word2Vec in **Python**
- Constructed Naïve Bayes Classifier model by **Scikit-Learn** and searched the best estimator by Grid Search in Python
- Exported and stored the result into **MySQL** database
- Worked with application development team to develop Web Service Application with **Java Spring Boot** and **Hibernate**.

Environments:

Apache NiFi, Spark, MongoDB, SPSS, Python, MySQL, AWS Kinesis, Lambda

Shanghai Pudong Development Bank, Shanghai, China

07/2015 - 06/2016

Project: Claims Management

Role: Data Analyst

Responsibilities:

- Tracked fluctuations in foreign exchange markets and transferred data from various Data Source using **Sqoop** to **HDFS** to perform analysis
- Created multiple **Hive** tables with partitioning and bucketing for efficient data access
- Developed time-series forecasting model for assets by **SAS**.
- Worked rigorously with **Impala** for executing ad-hoc queries
- Automated ETL process for large volume dataset by cleaning up the missing data, under-sampling the imbalanced dataset, and using filter-based feature selection and extraction in **Spark SQL** and **DataFrame API**.
- Wrote complex **SQL** queries to meet the business requirement in **MySQL**, **Oracle**, and **MS SQL**
- Developed **Java** scripts to maintain **MapReduce** job and database functions

EDUCATION:

- **Master of Science in Business Analytics**
- **Bachelor of Science in Mathematics**